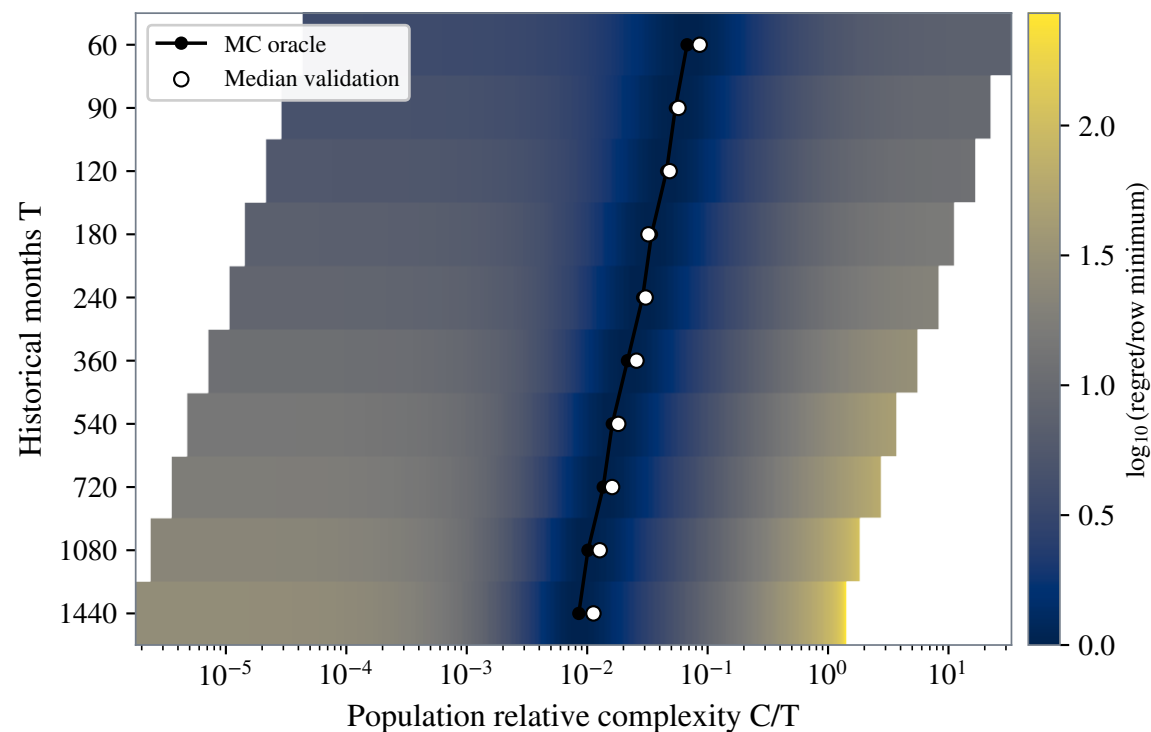


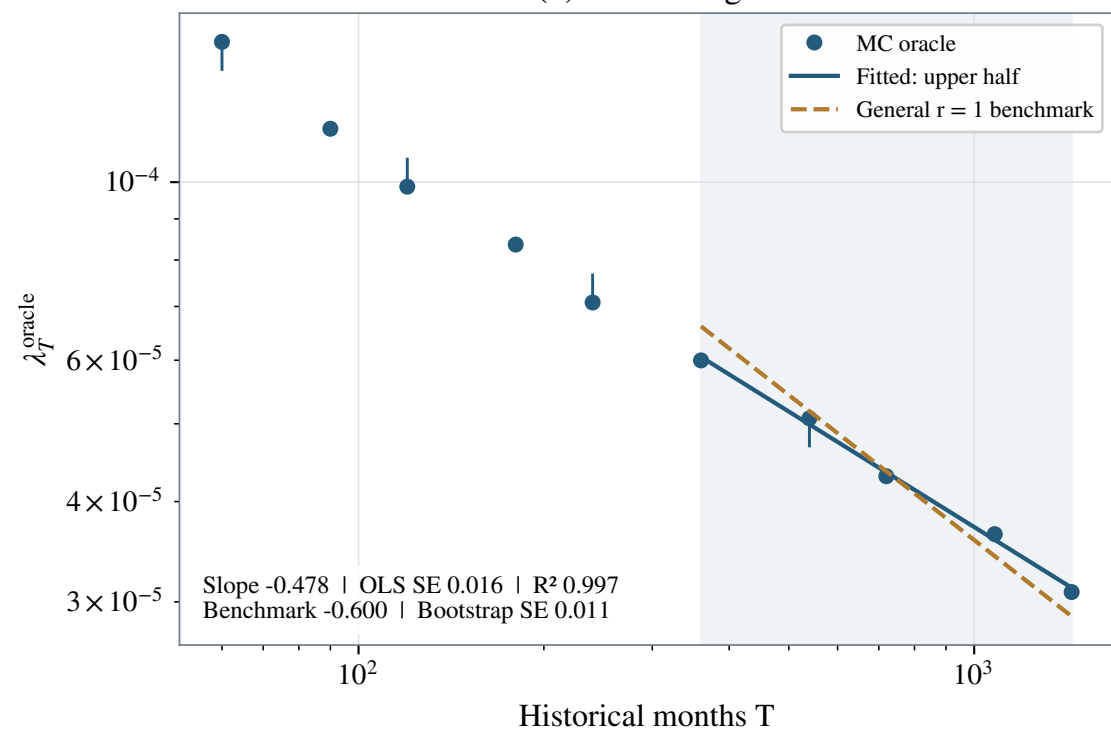
Learnability in a persistent characteristic-factor economy

Nonlinear policy | $b = 1.5$ | 5,000 stocks | 2,000 directions | 500 independent paths

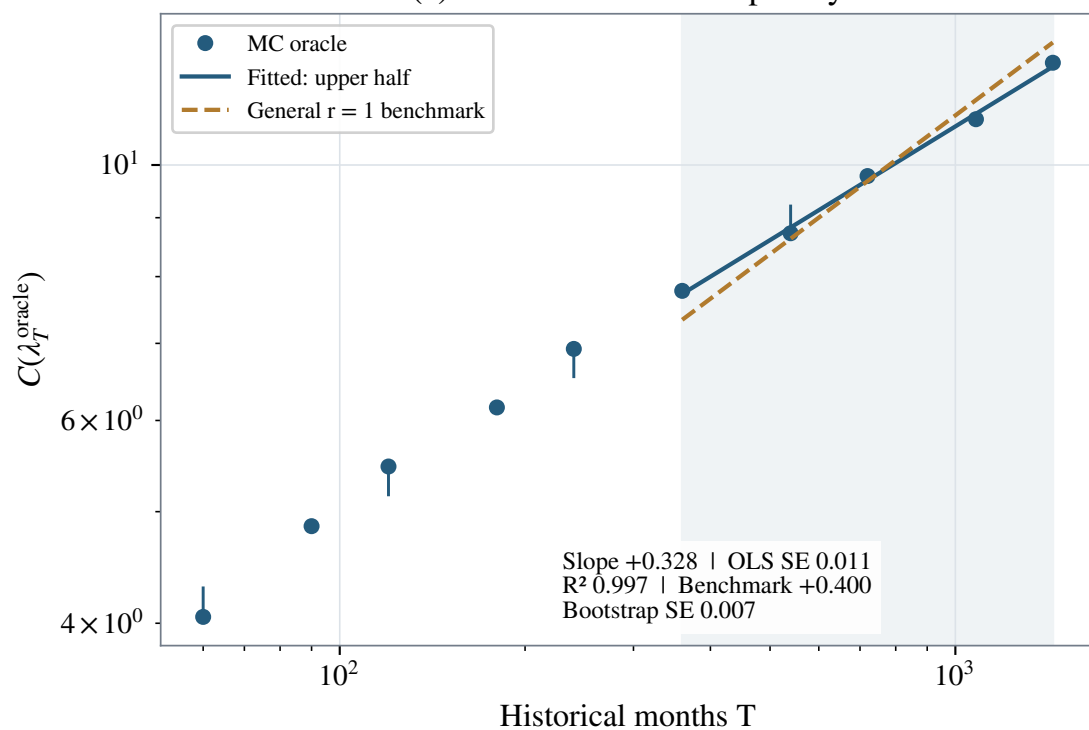
(a) Finite-sample learnability map



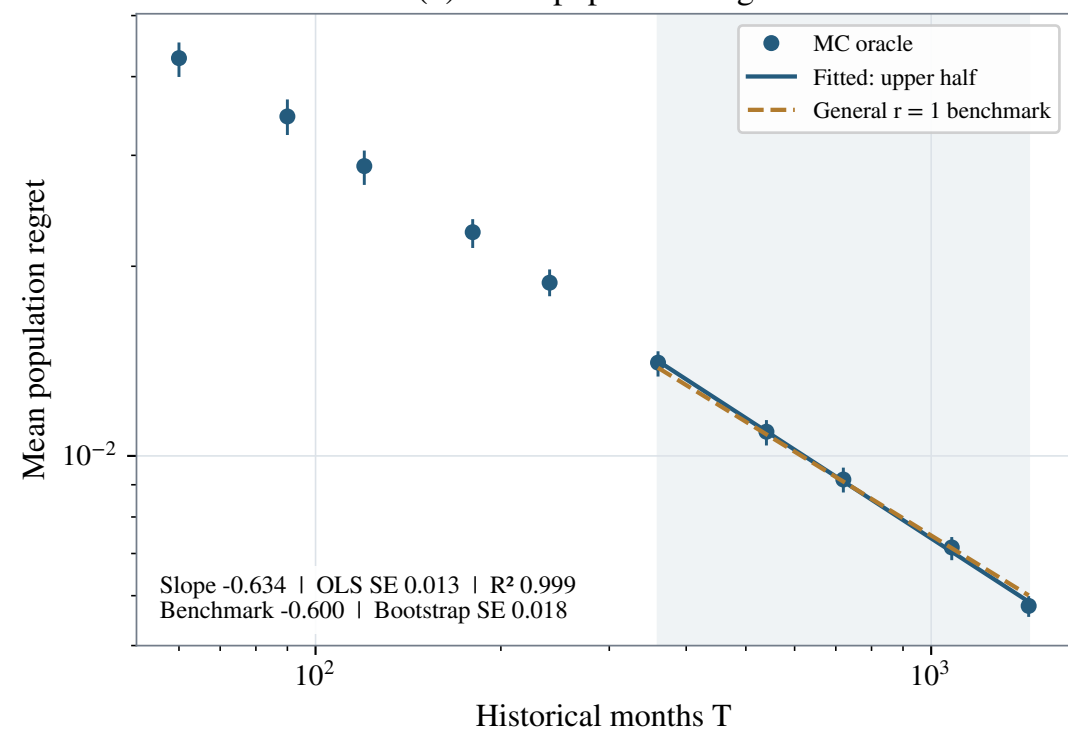
(b) Oracle ridge



(c) Oracle effective complexity



(d) Exact population regret



Risk is evaluated analytically, not on a simulated test set. Bars: 95% bootstrap intervals across independent paths. Slope fits use the predeclared upper half, $T = 360-1,440$ (shaded); all sample sizes remain visible. Dashed slopes describe the general RKHS envelope. The fixed smooth target need not attain its worst-case rate. The full C/T range is shown on a log axis and can exceed one. Median validation markers do not describe mean validation risk.